Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Index File

Student Name	**: Nosratee Jahan Naba**
Student ID	**: 1118005**
Course Title	: Operating System Sessional
Course Code	: CSE-312

Lab Report No	Lab Report Name	Submission Date	Total Marks	Obtained Marks
1	Installation of Linux	27/10/25	10	
2	Using Command	04/11/25	10	
3	Installation and use of LibraOffice	11/11/25	10	
4	Making Presentation using LibraOffice	09/12/25	10	
5	Introduction to Shell Scripting	16/12/25	10	
6	Advanced use of Shell Scripting	16/12/25	10	
7	Analysis of First Come First Serve (FCFS) Algorithm	23/12/25	10	
8	Analysis of SJF Algorithm	23/12/25	10	
9	Analysis of SRTF Algorithm	30/12/25	10	
10	Analysis of Round Robin Algorithm	30/12/25	10	

Total marks obtained :
Total number of lab reports :
Average marks :

Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

<u>Lab Report</u>

Lab Report No : 01

Lab Report Name : Installation of Linux.

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

ID : 1118005

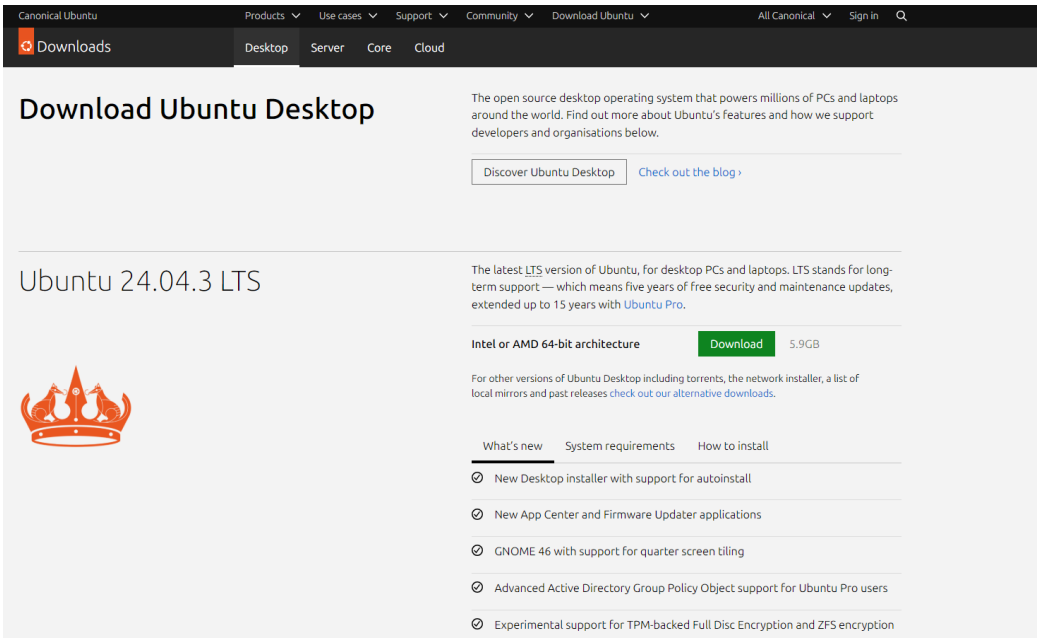
Level : 3 **Term** : I **Section** : A **Group** : G1

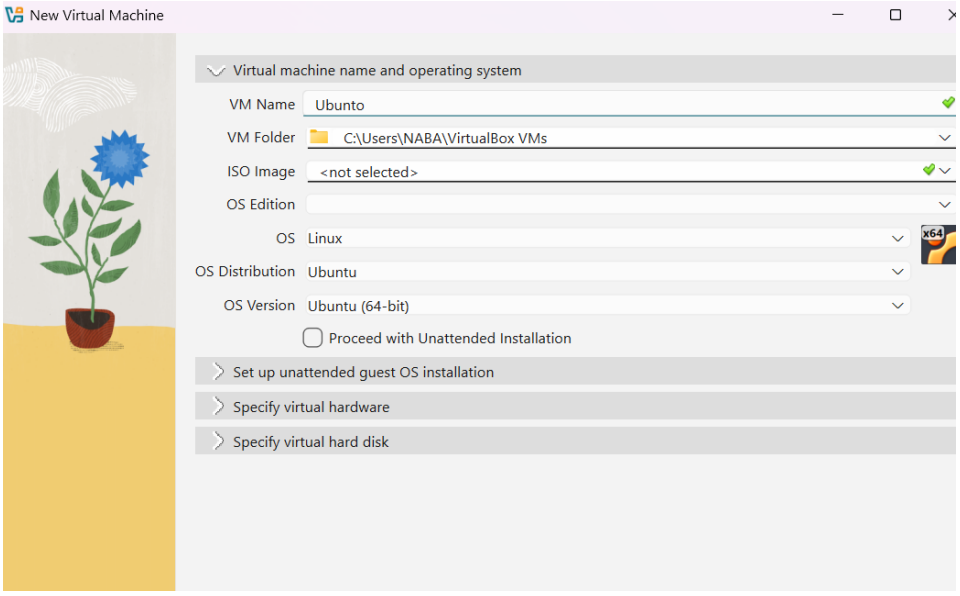
Date of Submission : 27/10/25 **Semester** : Fall **Year** : 2025

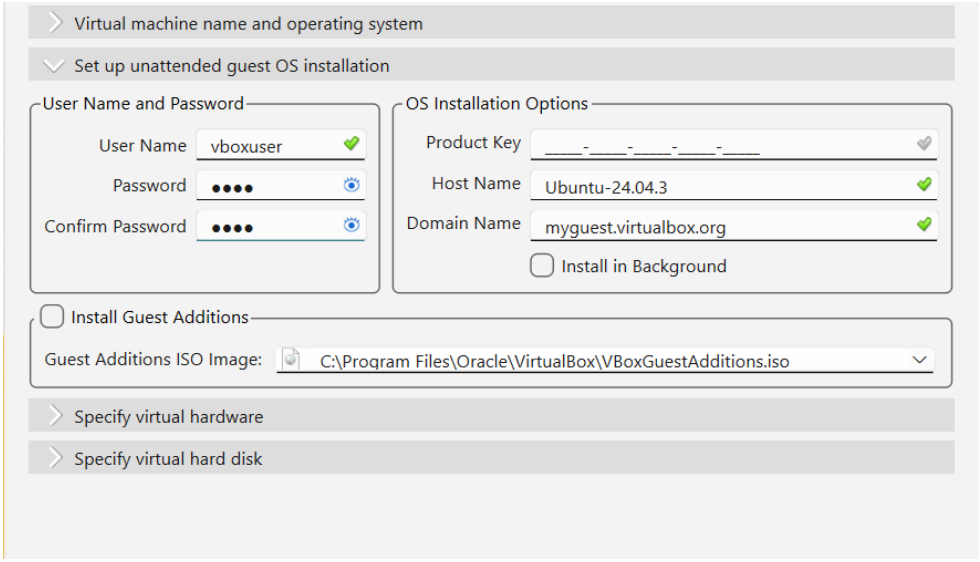
Key Learnings :

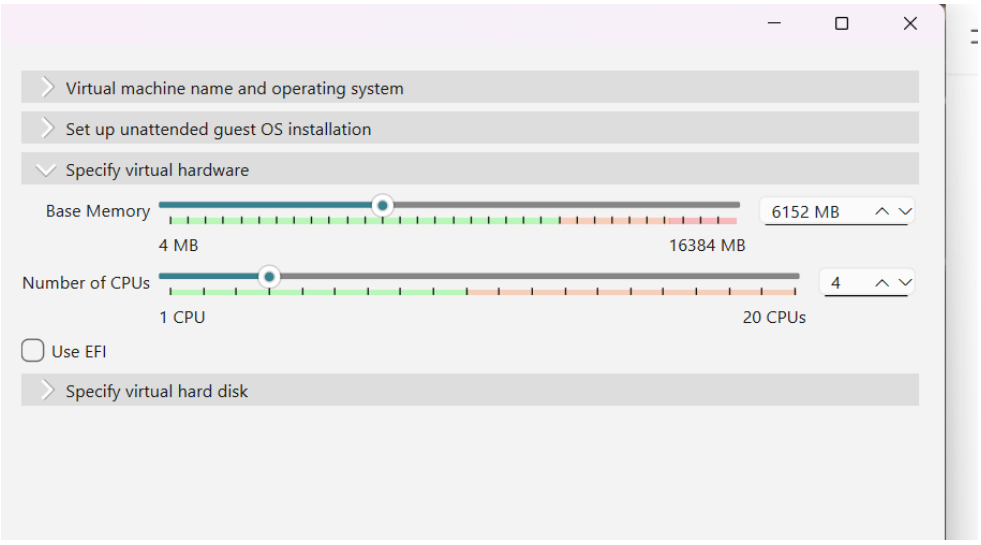
- Learned how to download Ubuntu Linux (Ubuntu 24.04 LTS) from the official website.
- Learned how to create and configure a virtual machine using Oracle VM VirtualBox.
- Understood the steps of setting OS type, version, memory, CPU, and storage for a virtual machine.
- Learned how to perform unattended installation by providing username, password, and system details
- Learned how to install Ubuntu Linux successfully inside a virtual machine environment.

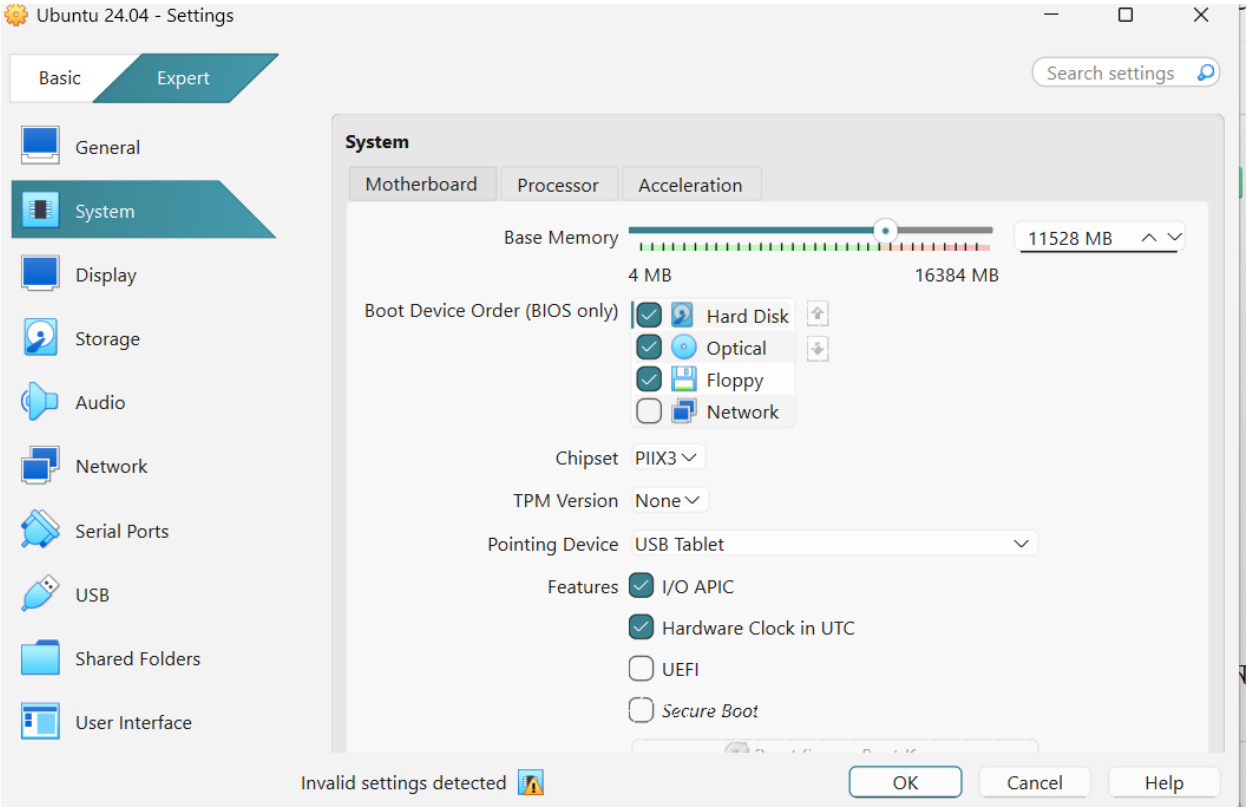
Installation of Ubuntu Linux Using Oracle VM VirtualBox :

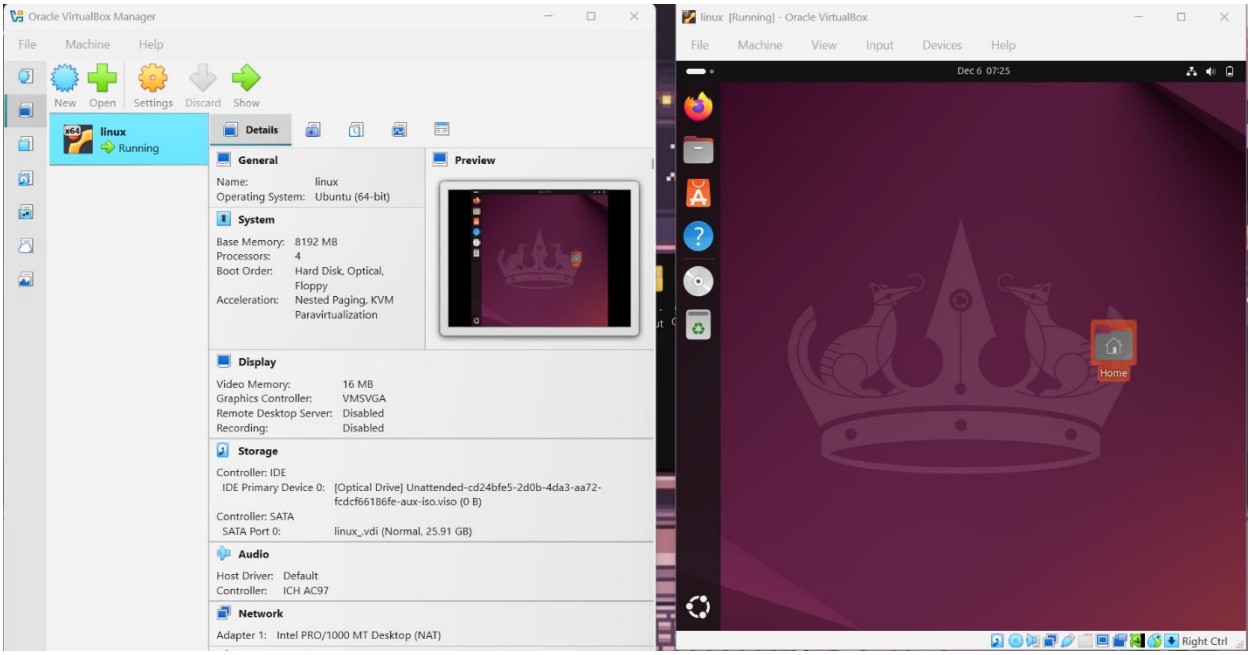



New Virtual Machine
Virtual machine name and operating system
VM Name
Ubunto
VM Folder
C:\Users\NABA\VirtualBox VMs
ISO Image
<not selected>
OS Edition
OS
Linux
OS Distribution
Ubuntu
OS Version
Ubuntu (64-bit)
Proceed with Unattended Installation
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk


Virtual machine name and operating system
Set up unattended guest OS installation
User Name and Password
User Name
vboxuser
Password
Confirm Password
OS Installation Options
Product Key
Host Name
Ubuntu-24.04.3
Domain Name
myguest.virtualbox.org
Install in Background
Install Guest Additions
Guest Additions ISO Image:
C:\Program Files\Oracle\VirtualBox\VBoxGuestAdditions.iso
Specify virtual hardware
Specify virtual hard disk


Virtual machine name and operating system
Set up unattended guest OS installation
Specify virtual hardware
Base Memory
6152 MB
4 MB
16384 MB
Number of CPUs
4
1 CPU
20 CPUs
Use EFI
Specify virtual hard disk


Ubuntu 24.04 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
System
Motherboard
Processor
Acceleration
Base Memory
11528 MB
4 MB
16384 MB
Boot Device Order (BIOS only)
Hard Disk
Optical
Floppy
Network
Chipset
PIIX3
TPM Version
None
Pointing Device
USB Tablet
Features
I/O APIC
Hardware Clock in UTC
UEFI
Secure Boot
Invalid settings detected
OK
Cancel
Help


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Show
linux
Running
Details
General
Name: linux
Operating System: Ubuntu (64-bit)
System
Base Memory: 8192 MB
Processors: 4
Boot Order: Hard Disk, Optical, Floppy
Acceleration: Nested Paging, KVM Paravirtualization
Preview
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Primary Device 0: [Optical Drive] Unattended-cd24bfe5-2d0b-4da3-aa72-fcdcf66186fe-aux-iso.viso (0 B)
Controller: SATA
SATA Port 0: linux_.vdi (Normal, 25.91 GB)
Audio
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
linux [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Dec 6 07:25
Home
Right Ctrl

Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 02

Lab Report Name : Using Commands.

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

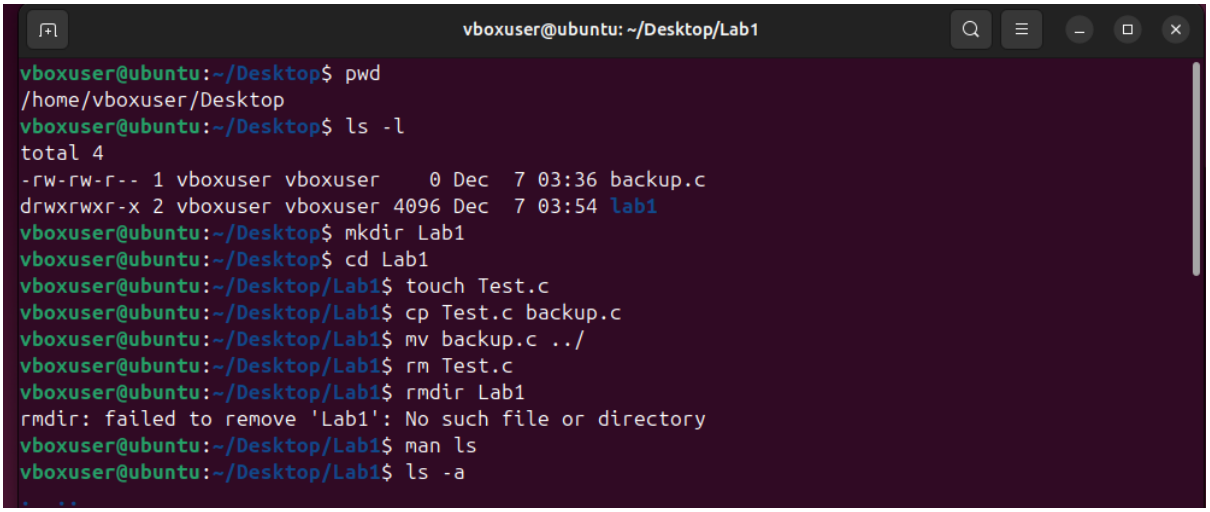
ID : 1118005

Level : 3 **Term** : I **Section** : A **Group** : G1

Date of Submission : 04/11/25 **Semester** : Fall **Year** : 2025

Key Learnings :

- Learned basic Linux navigation commands such as pwd, ls, and cd to move through directories.
- Learned how to create, copy, move, and delete files and directories using commands like mkdir, touch, cp, mv, and rm.
- Understood how file details and hidden files can be viewed using options like ls -l and ls -a.
- Learned how to view directory structure and search files using tree and find commands.
- Learned how to create and view file contents using redirection and commands like cat, head, and tail and many more.



```
vboxuser@ubuntu:~/Desktop/Lab1$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree  # version 2.1.3+pkg-5852, or
sudo apt  install tree  # version 2.1.1-2ubuntu3.24.04.2
See 'snap info tree' for additional versions.
vboxuser@ubuntu:~/Desktop/Lab1$ find . -name'*.c'
find: unknown predicate `-name*.c'
vboxuser@ubuntu:~/Desktop/Lab1$ cat>sample.txt
learning linux commands
^C
vboxuser@ubuntu:~/Desktop/Lab1$ head -5 sample.txt
learning linux commands
vboxuser@ubuntu:~/Desktop/Lab1$ tail -5 sample.txt
learning linux commands
vboxuser@ubuntu:~/Desktop/Lab1$ date
Sun Dec  7 07:14:30 AM UTC 2025
vboxuser@ubuntu:~/Desktop/Lab1$ whoami
vboxuser
```

```
vboxuser@ubuntu:~/Desktop/Lab1$ ps
    PID TTY          TIME CMD
   7225 pts/0    00:00:00 bash
   8346 pts/0    00:00:00 ps
vboxuser@ubuntu:~/Desktop/Lab1$ top

top - 07:14:44 up 20 min,  1 user,  load average: 0.18, 0.23, 0.29
Tasks: 235 total,   1 running, 234 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.7 us,  5.1 sy,  0.0 ni, 88.1 id,  5.1 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7941.8 total,   6065.1 free,   1237.2 used,    906.0 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   6704.6 avail Mem
```

```
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   8347 vboxuser  20   0   14536   5604   3404 R  15.4   0.1   0:00.04 top
   1247 root      20   0  295788   3864   3332 S   7.7   0.0   0:00.91 VBoxSer+
      1 root      20   0   23216  14172   9420 S   0.0   0.2   0:10.08 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.21 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     11 root       0 -20       0      0      0 I   0.0   0.0   0:00.15 kworker+
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker+
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.30 ksoftir+
vboxuser@ubuntu:~/Desktop/Lab1$ kill<pid>
bash: syntax error near unexpected token `newline'
vboxuser@ubuntu:~/Desktop/Lab1$ uptime
 07:15:07 up 20 min,  1 user,  load average: 0.27, 0.25, 0.29
```

```
vboxuser@ubuntu:~/Desktop/Lab1$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             39 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      4
  On-line CPU(s) list:       0-3
Vendor ID:                   GenuineIntel
  Model name:                13th Gen Intel(R) Core(TM) i7-13620H
    CPU family:              6
    Model:                   186
    Thread(s) per core:      1
    Core(s) per socket:      4
    Socket(s):               1
    Stepping:                2
    BogoMIPS:                5836.79
    Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pg
                             e mca cmov pat pse36 clflush mmx fxsr sse sse2 ht s
                             yscall nx rdtscp lm constant_tsc rep_good nopl xtop
                             ology nonstop_tsc cpuid tsc_known_freq pni pclmulqd
                             q ssse3 fma cx16 pcid sse4_1 sse4_2 movbe popcnt ae
                             s xsave avx f16c rdrand hypervisor lahf_lm abm 3dno
                             wprefetch ibrs_enhanced fsgsbase bmi1 avx2 bmi2 inv
                             pcid rdseed adx clflushopt sha_ni arat md_clear flu
                             sh_l1d arch_capabilities
Virtualization features:
  Hypervisor vendor:         KVM
  Virtualization type:       full
Caches (sum of all):
  L1d:                       192 KiB (4 instances)
  L1i:                       128 KiB (4 instances)
  L2:                        5 MiB (4 instances)
  L3:                        96 MiB (4 instances)
```

```
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-3
Vulnerabilities:
  Gather data sampling:      Not affected
  Ghostwrite:                Not affected
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:             Not affected
  L1tf:                      Not affected
  Mds:                       Not affected
  Meltdown:                  Not affected
  Mmio stale data:           Not affected
  Reg file data sampling:    Mitigation; Clear Register File
  Retbleed:                  Mitigation; Enhanced IBRS
  Spec rstack overflow:      Not affected
  Spec store bypass:         Vulnerable
  Spectre v1:                Mitigation; usercopy/swapgs barriers and __user poi
                             nter sanitization
  Spectre v2:                Mitigation; Enhanced / Automatic IBRS; PBRSB-eIBRS
                             SW sequence; BHI SW loop, KVM SW loop
  Srbds:                     Not affected
  Tsa:                       Not affected
  Tsx async abort:           Not affected
  Vmscape:                   Not affected
vboxuser@ubuntu:~/Desktop/Lab1$ uname -a
Linux ubuntu 6.14.0-36-generic #36~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Wed Oct 15 15:45:17 UTC 2 x
86_64 x86_64 x86_64 GNU/Linux
```

```
vboxuser@ubuntu:~/Desktop/Lab1$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           795M  1.5M  793M   1% /run
/dev/sda2        26G  7.0G   18G  29% /
tmpfs           3.9G     0  3.9G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           795M  128K  795M   1% /run/user/1000
/dev/sr0        6.0G  6.0G     0 100% /media/vboxuser/Ubuntu 24.04.3 LTS amd64
vboxuser@ubuntu:~/Desktop/Lab1$ du -sh*
du: invalid option -- '*'
Try 'du --help' for more information.
vboxuser@ubuntu:~/Desktop/Lab1$ du -sh *
4.0K    sample.txt
vboxuser@ubuntu:~/Desktop/Lab1$ free -m
               total        used        free      shared  buff/cache   available
Mem:            7941        1239        6043          34         925        6702
Swap:              0           0           0
```

Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 03

Lab Report Name : Installation and use of LibreOffice

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

ID : 1118005

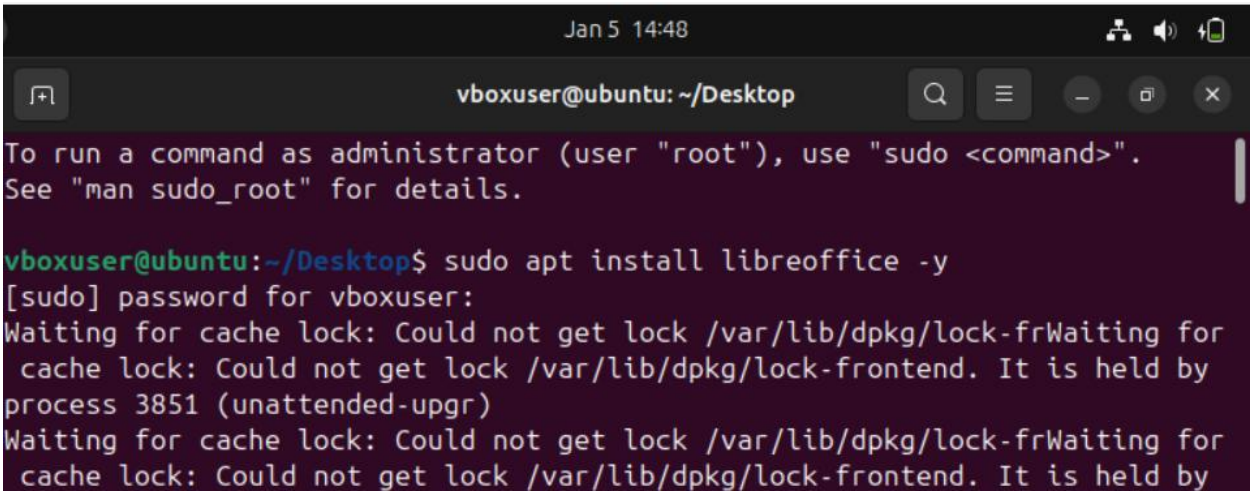
Level : 3 **Term** : I **Section** : A **Group** : G1

Date of Submission : 11/11/25 **Semester** : Fall **Year** : 2025

Key Learnings :

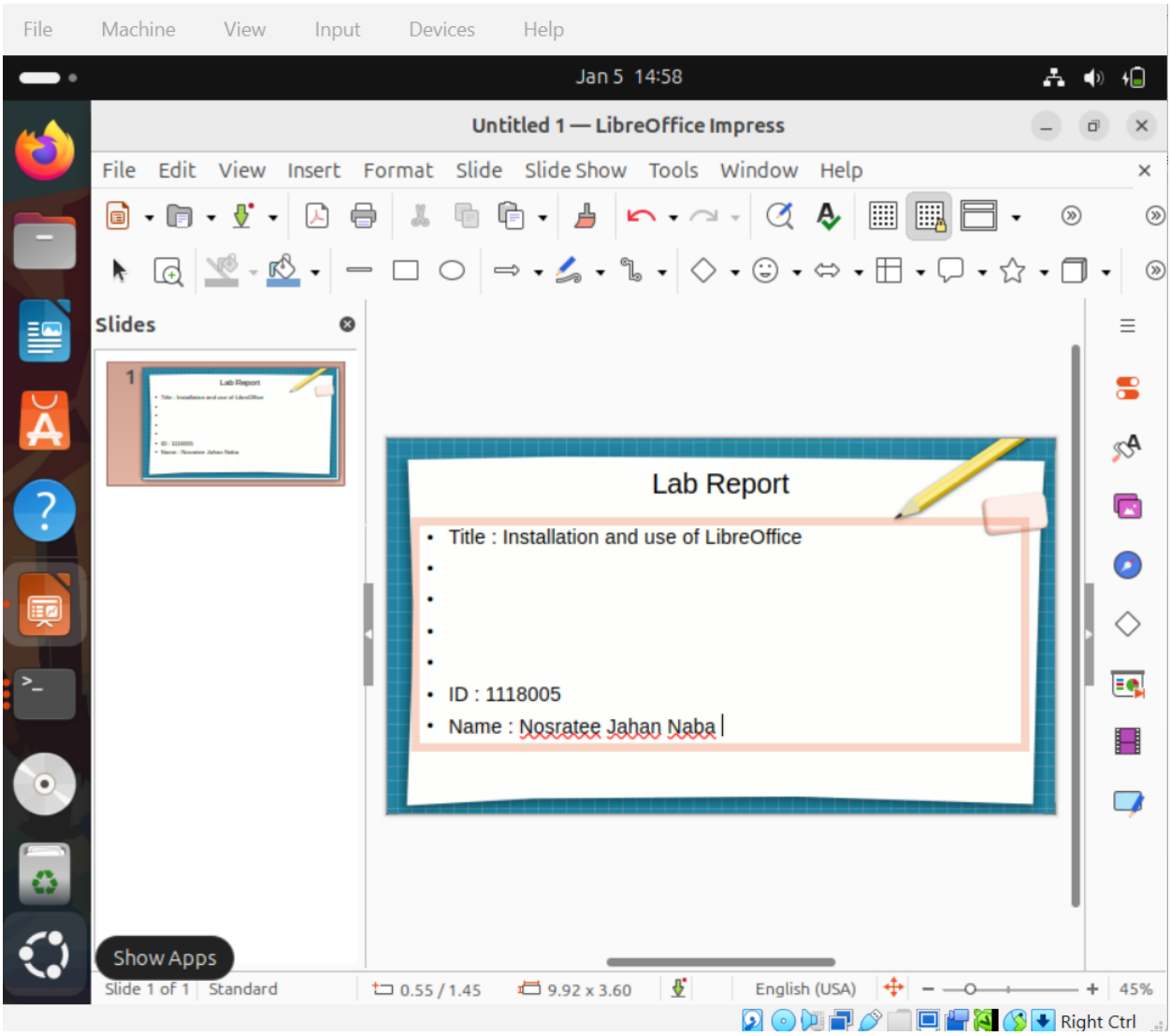
- Learned how to install LibreOffice on a computer.
- Learned how to open and navigate its applications
- Learned to create, edit, and save documents.
- Learned basic formatting and layout tools.
- Learned to use presentations for slides.

Installation:



```
Setting up openjdk-21-jre:amd64 (21.0.9+10-1~24.04) ...
Setting up libreoffice-script-provider-bsh (4:24.2.7-0ubuntu0.24
.04.4) ...
Setting up libreoffice-script-provider-js (4:24.2.7-0ubuntu0.24.
04.4) ...
Setting up libreoffice-sdbc-hsqldb (4:24.2.7-0ubuntu0.24.04.4) .
..
Setting up default-jre-headless (2:1.21-75+exp1) ...
Setting up libreoffice-wiki-publisher (4:1.2.0+LibO24.2.7-0ubunt
u0.24.04.4) ...
Setting up libreoffice-nlpsolver (4:0.9+LibO24.2.7-0ubuntu0.24.0
4.4) ...
Setting up default-jre (2:1.21-75+exp1) ...
vboxuser@ubuntu:~/Desktop$ libreoffice
```

Use of LibreOffice:



Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 04

Lab Report Name : Making Presentation Using LibreOffice

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

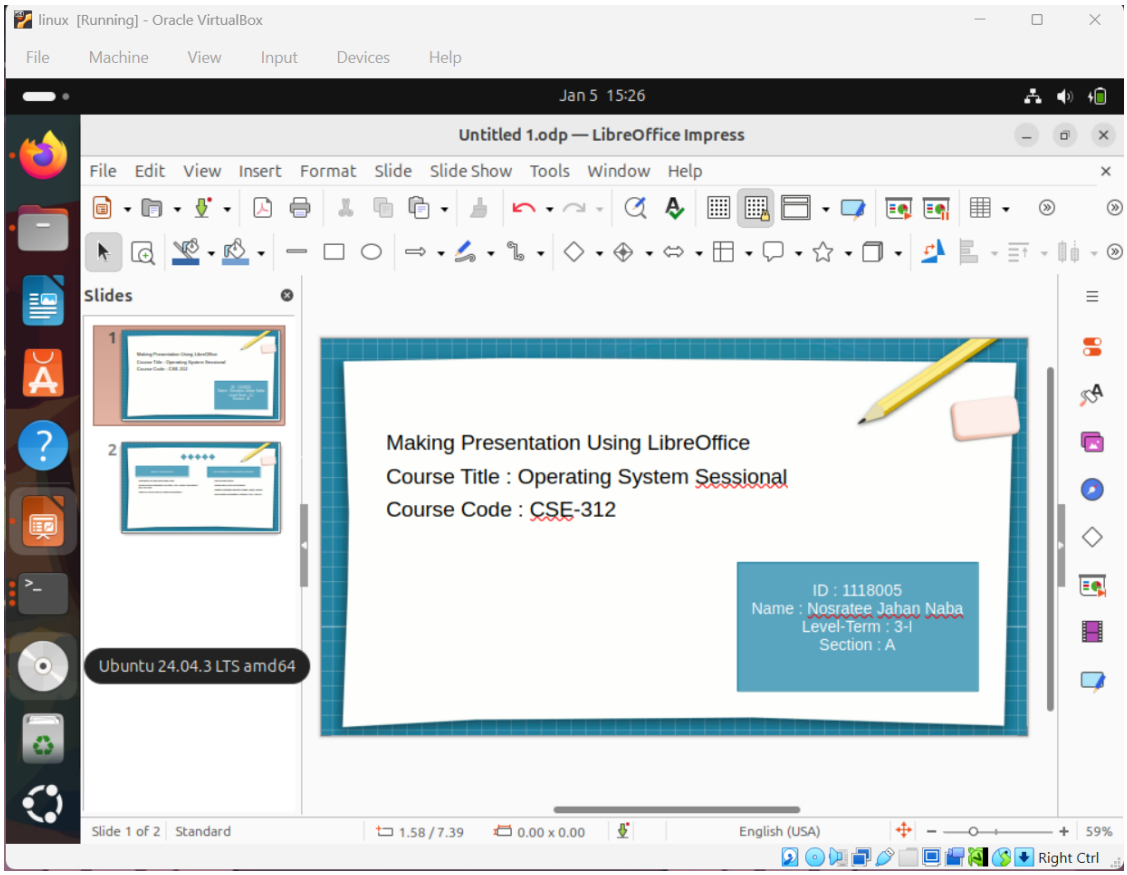
ID : 1118005

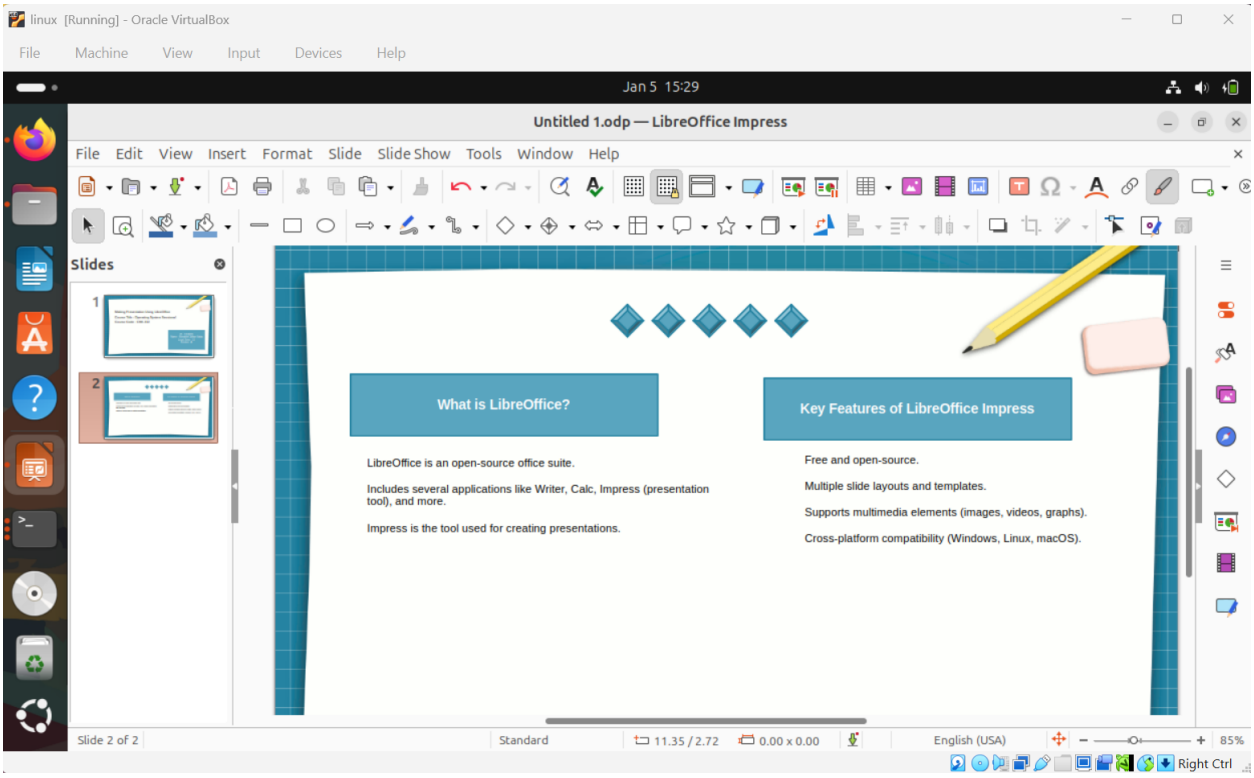
Level : 3 **Term** : I **Section** : A **Group** : G1

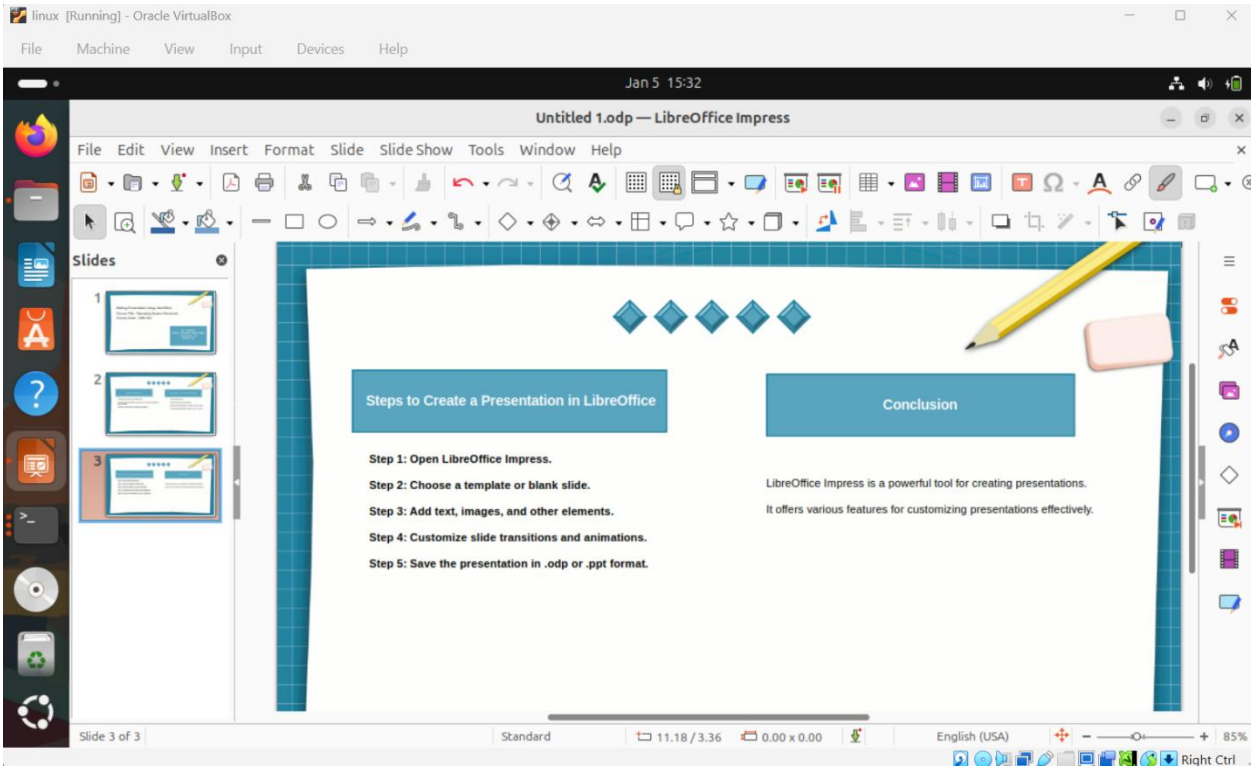
Date of Submission : 09/12/25 **Semester** : Fall **Year** : 2025

Key Learnings :

- Learned how to open LibreOffice Impress to create presentations.
- Learned to add slides, text, and images.
- Learned to format text, apply themes, and design layouts.
- Learned to run slideshow and save/export presentations.




linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 5 15:29
Untitled 1.odp — LibreOffice Impress
File Edit View Insert Format Slide Slide Show Tools Window Help
Slides
What is LibreOffice?
LibreOffice is an open-source office suite.
Includes several applications like Writer, Calc, Impress (presentation tool), and more.
Impress is the tool used for creating presentations.
Key Features of LibreOffice Impress
Free and open-source.
Multiple slide layouts and templates.
Supports multimedia elements (images, videos, graphs).
Cross-platform compatibility (Windows, Linux, macOS).
Slide 2 of 2
Standard
11.35 / 2.72
0.00 x 0.00
English (USA)
85%
Right Ctrl


linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jan 5 15:32
Untitled 1.odp — LibreOffice Impress
File Edit View Insert Format Slide Slide Show Tools Window Help
Slides
Steps to Create a Presentation in LibreOffice
Step 1: Open LibreOffice Impress.
Step 2: Choose a template or blank slide.
Step 3: Add text, images, and other elements.
Step 4: Customize slide transitions and animations.
Step 5: Save the presentation in .odp or .ppt format.
Conclusion
LibreOffice Impress is a powerful tool for creating presentations.
It offers various features for customizing presentations effectively.
Slide 3 of 3
Standard
11.18 / 3.36
0.00 x 0.00
English (USA)
85%
Right Ctrl

Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 05

Lab Report Name : Introduction to Shell Scripting

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

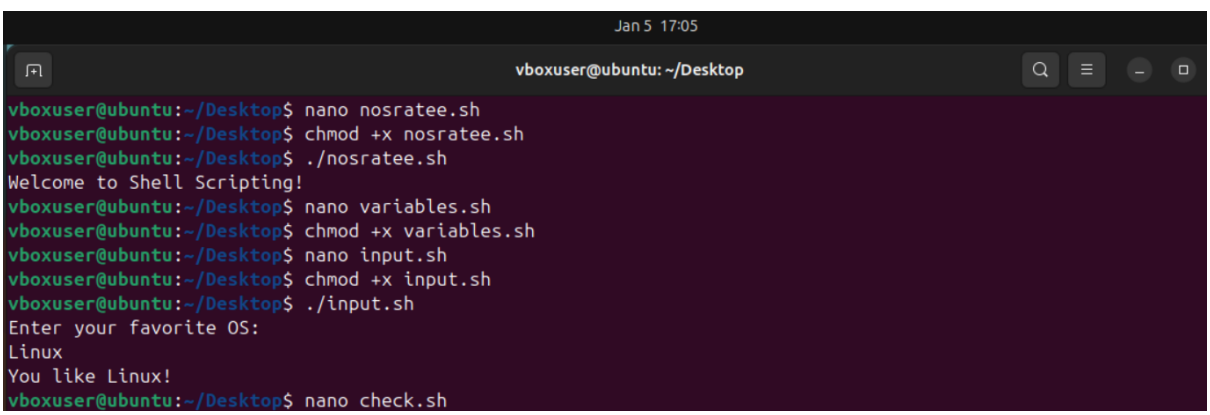
ID : 1118005

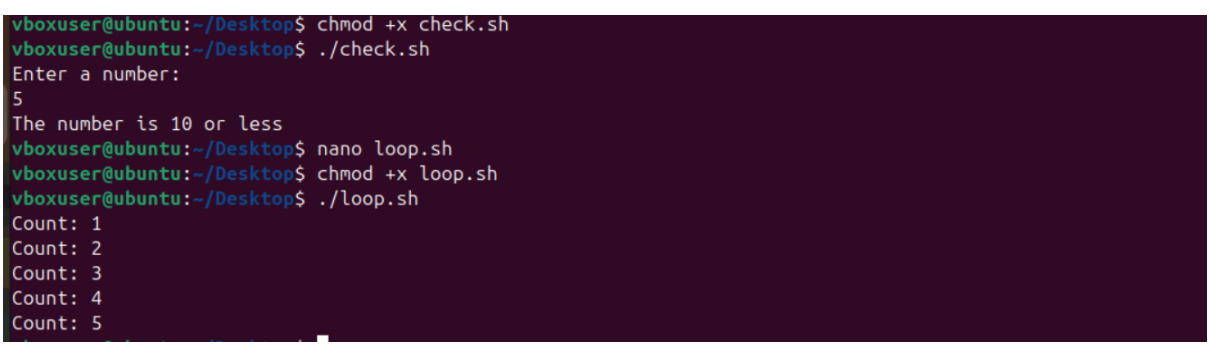
Level : 3 **Term** : I **Section** : A **Group** : G1

Date of Submission : 16/12/25 **Semester** : Fall **Year** : 2025

Key Learnings :

- Learned to create, save, and run shell scripts.
- Learned variables, user input, and echo output.
- Learned if-else conditions and loops.
- Learned chmod +x to make scripts executable.





Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 06

Lab Report Name : Advance use of Shell Scripting

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

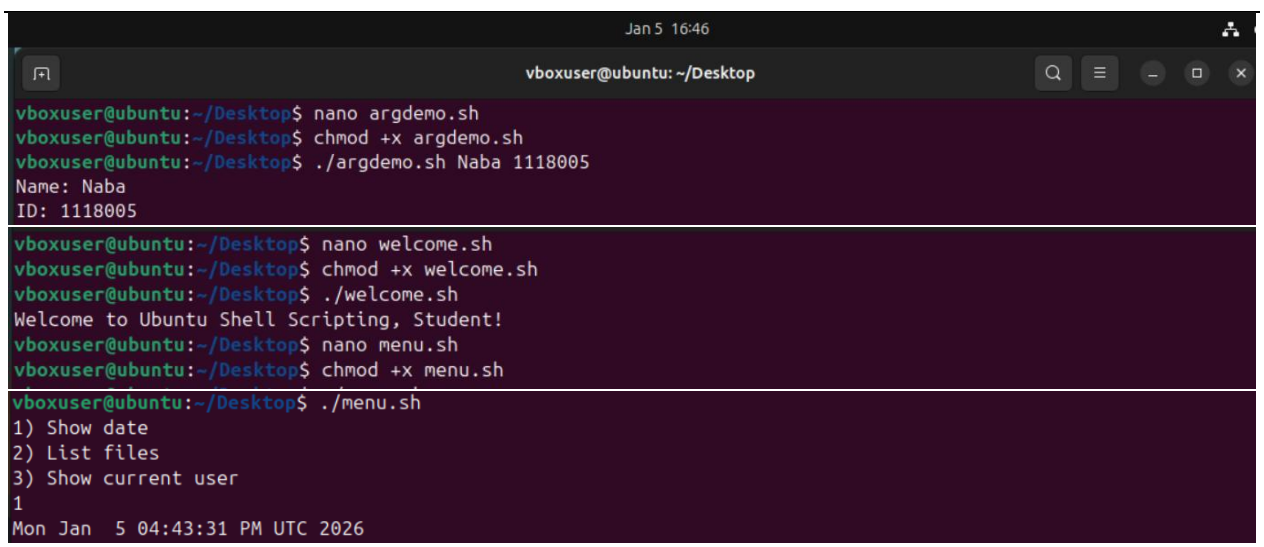
ID : 1118005

Level : 3 **Term** : I **Section** : A **Group** : G1

Date of Submission : 16/12/25 **Semester** : Fall **Year** : 2025

Key Learnings :

- Learned to pass arguments to scripts and access them with $1, $2, etc.
- Learned to create and use functions in shell scripts.
- Learned to make interactive menus using case statements.
- Learned to loop through files and perform operations on each file.
- Learned to generate simple system reports (RAM, CPU, disk, date).



```
vboxuser@ubuntu:~/Desktop$ nano loopfiles.sh
vboxuser@ubuntu:~/Desktop$ chmod +x loopfiles.sh
vboxuser@ubuntu:~/Desktop$ ./loopfiles.sh
./loopfiles.sh: line 7: *.txt: No such file or directory
File: *.txt has  lines
vboxuser@ubuntu:~/Desktop$ touch file1.txt file2.txt
vboxuser@ubuntu:~/Desktop$ nano sysreport.sh
vboxuser@ubuntu:~/Desktop$ chmod +x sysreport.sh
vboxuser@ubuntu:~/Desktop$ ./sysreport.sh
System Report

Date: Mon Jan  5 04:45:35 PM UTC 2026
User: vboxuser
Memory:
               total        used        free      shared  buff/cache   available
Mem:           7.8Gi       2.6Gi       780Mi        59Mi       4.4Gi       5.2Gi
Swap:             0B          0B          0B
Disk:
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           795M  1.7M  793M   1% /run
/dev/sda2        26G  9.1G   15G  38% /
tmpfs           3.9G     0  3.9G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           795M  156K  795M   1% /run/user/1000
/dev/sr0        6.0G  6.0G     0 100% /media/vboxuser/Ubuntu 24.04.3 LTS amd64
```

Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 07

Lab Report Name : Analysis of First Come First Serve (FCFS) Scheduling

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

ID : 1118005

Level : 3 **Term** : I **Section** : A **Group** : G1

Date of Submission : 23/12/25 **Semester** : Fall **Year** : 2025

Key Learnings :

- Learned FCFS scheduling executes processes in arrival order.
- Learned how to calculate Waiting Time (WT) and Turnaround Time (TAT).
- Learned how to compute average WT and TAT.
- Learned that short processes may wait long (convoy effect).

Code Implementation :

Programiz C++ Online Compiler

main.cpp Share Run

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cout << "Enter Number of Processes: ";
    cin >> n;

    int burst_time[n], waiting_time[n], turnaround_time[n];

    cout << "Enter Burst time for each Process:\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i + 1 << ": ";
        cin >> burst_time[i];
    }

```

```
    waiting_time[0] = 0;
    for (int i = 1; i < n; i++) {
        waiting_time[i] = waiting_time[i - 1] + burst_time[i - 1];
    }

    for (int i = 0; i < n; i++) {
        turnaround_time[i] = waiting_time[i] + burst_time[i];
    }

    cout << "\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i + 1 << "\t\t\t"
             << burst_time[i] << "\t\t\t"
             << waiting_time[i] << "\t\t\t"
             << turnaround_time[i] << endl;
    }
    return 0;
}
```

Output

```
Enter Number of Processes: 3
Enter Burst time for each Process:
P1: 5
P2: 3
P3: 8

Process Burst Time  Waiting Time    Turnaround Time
P1          5           0           5
P2          3           5           8
P3          8           8           16


=== Code Execution Successful ===
```

Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 08

Lab Report Name : Analysis of Shortest Job First (SJF) Scheduling

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

ID : 1118005

Level : 3 **Term** : I **Section** : A **Group** : G1

Date of Submission : 23/12/25 **Semester** : Fall **Year** : 2025

Key Learnings :

- **Learned SJF scheduling executes the shortest burst time process first.**
- **Learned how to calculate Waiting Time (WT) and Turnaround Time (TAT) for SJF.**
- **Learned how to compute average WT and TAT.**
- **Learned that SJF reduces average waiting time compared to FCFS.**
- **Learned the possibility of starvation for long processes.**

Code Implementation :

Programiz C++ Online Compiler

main.cpp Share Run

```cpp
#include <bits/stdc++.h>
using namespace std;

int main()
{
    int n;
    cout << "Enter Number of Processes: ";
    cin >> n;

    int bt[n], wt[n], tat[n], pid[n];
    float avg_wt = 0, avg_tat = 0;

    cout << "Enter Burst Time for each Process:\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i + 1 << ": ";
        cin >> bt[i];
        pid[i] = i + 1;
    }
```

```cpp
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                int temp_bt = bt[i];
                bt[i] = bt[j];
                bt[j] = temp_bt;

                int temp_pid = pid[i];
                pid[i] = pid[j];
                pid[j] = temp_pid;
            }
        }
    }
    wt[0] = 0;
    for (int i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for (int i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
```

```cpp
    cout << "\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << pid[i] << "\t\t\t"
             << bt[i] << "\t\t\t\t"
             << wt[i] << "\t\t\t\t"
             << tat[i] << endl;
    }
    float total_wt = 0, total_tat = 0;
    for (int i = 0; i < n; i++) {
        total_wt += wt[i];
        total_tat += tat[i];
    }

    avg_wt = total_wt / n;
    avg_tat = total_tat / n;

    cout << fixed << setprecision(2);
    cout << "\nAverage Waiting Time: " << avg_wt << endl;
    cout << "Average Turnaround Time: " << avg_tat << endl;
    return 0;
}
```

Output

```
Enter Number of Processes: 3
Enter Burst Time for each Process:
P1: 5
P2: 3
P3: 4

Process Burst Time  Waiting Time    Turnaround Time
P2          3               0               3
P3          4               3               7
P1          5               7               12

Average Waiting Time: 3.33
Average Turnaround Time: 7.33


=== Code Execution Successful ===
```

Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 09

Lab Report Name : Analysis of Shortest Remaining Time First (SRTF) Scheduling

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

ID : 1118005

Level : 3 **Term** : I **Section** : A **Group** : G1

Date of Submission : 30/12/25 **Semester** : Fall **Year** : 2025

Key Learnings :

- Learned SRTF scheduling is preemptive, always runs the process with shortest remaining burst time.
- Learned how to calculate Waiting Time (WT) and Turnaround Time (TAT) in preemptive scheduling.
- Learned how to compute average WT and TAT.
- Learned it minimizes average waiting time compared to FCFS and non-preemptive SJF.
- Learned long processes may suffer starvation if short processes keep arriving.

Code Implementation:

Programiz C++ Online Compiler

main.cpp Share Run

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cout << "Enter number of processes: ";
    cin >> n;

    vector<int> bt(n), at(n), wt(n), tat(n), rt(n);
    int completed = 0;
    float avg_wt = 0, avg_tat = 0;

    cout << "Enter Arrival Time and Burst Time for each process:\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i+1 << " Arrival Time: ";
        cin >> at[i];
        cout << "P" << i+1 << " Burst Time: ";
        cin >> bt[i];
        rt[i] = bt[i];
    }
```

```
    int time = 0;
    while (completed < n) {
        int idx = -1;
        int min_rt = INT_MAX;
        for (int i = 0; i < n; i++) {
            if (at[i] <= time && rt[i] > 0 && rt[i] < min_rt) {
                min_rt = rt[i];
                idx = i;
            }
        }
        if (idx != -1) {
            rt[idx]--;
            time++;
            if (rt[idx] == 0) {
                completed++;
                tat[idx] = time - at[idx];
                wt[idx] = tat[idx] - bt[idx];
            }
        } else {
            time++;
        }
    }
```

```
    cout << "\nProcess\tAT\tBT\tWT\tTAT\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i+1 << "\t"
             << at[i] << "\t\t\t"
             << bt[i] << "\t\t\t"
             << wt[i] << "\t\t\t"
             << tat[i] << endl;
        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    avg_wt /= n;
    avg_tat /= n;

    cout << fixed << setprecision(2);
    cout << "\nAverage Waiting Time: " << avg_wt << endl;
    cout << "Average Turnaround Time: " << avg_tat << endl;

    return 0;
}
```

Output

```
Enter number of processes: 3
Enter Arrival Time and Burst Time for each process:
P1 Arrival Time: 0
P1 Burst Time: 7
P2 Arrival Time: 2
P2 Burst Time: 4
P3 Arrival Time: 4
P3 Burst Time: 1

Process AT  BT  WT  TAT
P1  0           7           5           12
P2  2           4           1           5
P3  4           1           0           1

Average Waiting Time: 2.00
Average Turnaround Time: 6.00
```

Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 10

Lab Report Name : Analysis of Round Robin Scheduling

Course Title : Operating System Sessional.

Course Code : CSE 312

Name : Nosratee Jahan Naba

ID : 1118005

Level : 3 **Term** : I **Section** : A **Group** : G1

Date of Submission : 30/12/25 **Semester** : Fall **Year** : 2025

Key Learnings :

- Learned Round Robin scheduling is preemptive and uses a time quantum for each process.
- Learned how to calculate Waiting Time (WT) and Turnaround Time (TAT) with multiple CPU bursts.
- Learned how to compute average WT and TAT.
- Learned fair CPU sharing among all processes.
- Learned smaller time quantum → more context switching, larger quantum → behaves like FCFS.

Code Implementation:

Programiz C++ Online Compiler

main.cpp Share Run

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n, tq;
    cout << "Enter number of processes: ";
    cin >> n;
    cout << "Enter Time Quantum: ";
    cin >> tq;

    int bt[n], at[n], rt[n], wt[n], tat[n];
    for(int i=0; i<n; i++) {
        cout << "P" << i+1 << " Arrival Time: ";
        cin >> at[i];
        cout << "P" << i+1 << " Burst Time: ";
        cin >> bt[i];
        rt[i] = bt[i];
        wt[i] = 0;
    }

```

```
    int time = 0, completed = 0;
    while(completed < n) {
        bool done = true;
        for(int i=0; i<n; i++) {
            if(rt[i] > 0 && at[i] <= time) {
                done = false;
                if(rt[i] > tq) {
                    rt[i] -= tq;
                    time += tq;
                } else {
                    time += rt[i];
                    wt[i] = time - at[i] - bt[i];
                    rt[i] = 0;
                    completed++;
                }
            }
        }
        if(done) time++;
    }

    for(int i=0; i<n; i++) {
        tat[i] = bt[i] + wt[i];
    }

    float avg_wt = 0, avg_tat = 0;
    cout << "\nProcess\tAT\tBT\tWT\tTAT\n";
    for(int i=0; i<n; i++) {
        cout << "P" << i+1 << "\t"
             << at[i] << "\t"
             << bt[i] << "\t"
             << wt[i] << "\t"
             << tat[i] << endl;
        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    avg_wt /= n;
    avg_tat /= n;

    cout << fixed << setprecision(2);
    cout << "\nAverage Waiting Time: " << avg_wt << endl;
    cout << "Average Turnaround Time: " << avg_tat << endl;

    return 0;
}
```

Output

```
Enter number of processes: 3
Enter Time Quantum: 2
P1 Arrival Time: 0
P1 Burst Time: 5
P2 Arrival Time: 1
P2 Burst Time: 3
P3 Arrival Time: 2
P3 Burst Time: 1

Process AT  BT  WT  TAT
P1  0   5   4   9
P2  1   3   4   7
P3  2   1   2   3

Average Waiting Time: 3.33
Average Turnaround Time: 6.33


=== Code Execution Successful ===
```